

Improving Accuracy and Calibration via Differentiated Deep Mutual Learning

Han Liu^{1,2,†}, Peng Cui^{1,†}, Binging Wang², Weipeng Chen², Yupeng Zhang²,
Jun Zhu^{1,*}, and Xiaolin Hu^{1,3,4,*}

¹Department of Computer Science and Technology, Institute for AI,
BNRist, Tsinghua University, Beijing 100084, China

²Baichuan Inc.

³Tsinghua Laboratory of Brain and Intelligence (THBI),
IDG/McGovern Institute for Brain Research, Tsinghua University, Beijing 100084, China

⁴Chinese Institute for Brain Research (CIBR), Beijing 100010, China

{han-liu18, cuip22}@emails.tsinghua.edu.cn

{daniel, chenweipeng, zhangyupeng}@baichuan-inc.com

{dcszj, xlhu}@mail.tsinghua.edu.cn

Abstract

Deep Neural Networks (DNNs) have achieved remarkable success in a variety of tasks, particularly in terms of prediction accuracy. However, in real-world scenarios, especially in safety-critical applications, accuracy alone is insufficient; reliable uncertainty estimates are essential. Modern DNNs, often trained with cross-entropy loss, tend to exhibit overconfidence, especially on ambiguous samples. Many techniques aim to improve uncertainty calibration, yet they often come at the cost of reduced accuracy or increased computational demands. To address this challenge, we propose Differentiated Deep Mutual Learning (Diff-DML), an efficient ensemble approach that simultaneously enhances accuracy and uncertainty calibration. Diff-DML draws inspiration from Deep Mutual Learning (DML) while introducing two strategies to maintain prediction diversity: (1) Differentiated Training Strategy (DTS) and (2) Diversity-Preserving Learning Objective (DPLO). Our theoretical analysis shows that Diff-DML's diversified learning framework not only leverages ensemble benefits but also avoids the loss of prediction diversity observed in traditional DML setups, which is crucial for improved calibration. Extensive evaluations on various benchmarks confirm the effectiveness of Diff-DML. For instance, on the CIFAR-100 dataset, Diff-DML on ResNet34/50 models achieved substantial improvements over the previous state-of-the-art method, MDCA, with absolute accuracy gains of 1.3%/3.1%, relative ECE reductions

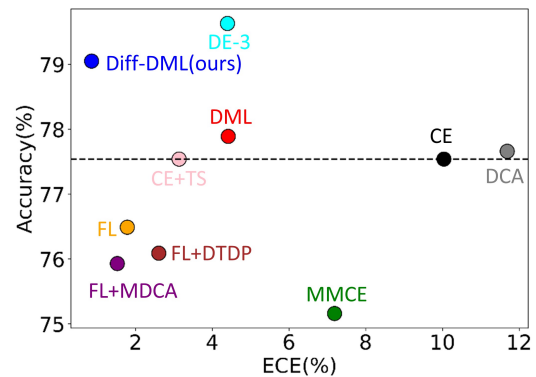


Figure 1. Scatter plot of ECE and accuracy for different calibration methods based on the ResNet50 backbone on the CIFAR-100 test set. Methods located closer to the top left corner perform better.

of 49.6%/43.8%, and relative classwise-ECE reductions of 7.7%/13.0%.

1. Introduction

Deep Neural Networks (DNNs) have achieved great success in various tasks, especially in terms of prediction accuracy. In safety-critical areas such as medical diagnosis [4, 18], self-driving cars [20], and protein engineering [6], accurate uncertainty estimation (i.e., well-calibrated prediction) is crucial for reliable model deployment. For instance, a reliable model needs to show high uncertainty when making incorrect predictions. However, models trained with cross-entropy (CE) loss tend to be overconfident, especially when

This work was supported by the National Key Research and Development Program of China (No. 2021ZD0200301) and the National Natural Science Foundation of China (No. U2341228).

[†]Equal contribution. ^{*}Corresponding author.

encountering ambiguous samples, leading to predictions that are often close to 100% confident and producing undesirable overconfidence [2, 8].

To address this issue, various regularization techniques have been explored during training, referred to as *training-time calibration*. For instance, entropy regularization (ER) [26] and label smoothing (LS) [22, 28] promote smoother predictive probability distributions to alleviate overconfidence by maximizing entropy or adjusting label distributions. Some methods aim to directly reduce calibration error during training to achieve multi-class calibration. For example, MMCE [16] introduces a trainable measure based on kernel mean embeddings, while MDCA [10] employs a differentiable auxiliary loss to minimize the gap between prediction confidence and accuracy. However, despite mitigating overconfidence, these techniques often compromise prediction accuracy due to excessive regularization. Another category of methods is post-processing, which adjusts output probabilities using Temperature Scaling (TS) [27], Vector Scaling (VS) [27], or Dirichlet Scaling (DS) [15]. However, these post-processing techniques typically rely on an additional validation dataset. Furthermore, they have limited learnable parameters, which may lead to suboptimal calibration performance [10].

Deep Ensembles (DE) [5, 17] offer a straightforward solution by combining multiple randomly initialized trained DNNs to improve prediction accuracy and uncertainty calibration. Yet, using multiple DNNs during training and inference significantly increases computational costs, which poses challenges for deploying DE in real-world scenarios. To address this limitation, we draw inspiration from Deep Mutual Learning (DML) [29], a technique used in model distillation, where an ensemble of student models learn collaboratively and teach each other throughout the training process. This work provides the theoretical analysis showing that DML encourages each model to learn from an ensemble distribution through a mimicry loss. The proposed theory motivates us to leverage DML as an implicit form of DE, aiming to achieve similar gains in accuracy and calibration as DE, while mitigating the high computational overhead because DML uses only one model for prediction during inference.

However, in standard DML, although the models start from different initial states, they become overly similar during mutual learning, which reduces the diversity of their outputs—a key factor needed for effective ensemble-based calibration. In this work, we propose *Differentiated Deep Mutual Learning (Diff-DML)*, a novel approach that enhances the benefits of DML by promoting prediction diversity between models to achieve high accuracy and strong calibration performance. Diff-DML incorporates two key strategies that preserve output diversity: (1) *Differentiated Training Strategy (DTS)*, which uses different learning rates and schedules

for different models to prevent premature convergence, and (2) *Diversity-Preserving Learning Objective (DPLO)*, which removes cross-entropy loss for auxiliary models to avoid overfitting to the one-hot ground-truth labels.

These strategies help Diff-DML balance collaborative learning with prediction diversity, resulting in better uncertainty calibration and enhanced predictive performance. Extensive evaluations on various benchmarks, including image classification, misclassification detection, and out-of-distribution (OOD) detection, confirm Diff-DML's effectiveness. For example, on the CIFAR-100 dataset, Diff-DML on ResNet34/50 achieved significant improvements compared to the previous state-of-the-art method, MDCA [10], with absolute accuracy increases of 1.3%/3.1%, relative reductions in ECE of 49.6%/43.8%, and relative reductions in classwise-ECE of 7.7%/13.0%. The latter two metrics measure calibration performance, where lower values indicate better results.

Figure 1 displays ECE and accuracy of different calibration methods, showing Diff-DML achieving favorable calibration and accuracy. Notably, Diff-DML incurs only minimal computational cost at inference, unlike DE, which requires multiple models to generate a single prediction.

2. Background and related work

Calibration, the process of ensuring that the predicted probabilities reflect the true likelihood of the events, plays a crucial role in machine learning models. It ensures that the prediction of machine learning models are not only accurate but also well-calibrated. Well-calibrated models provide reliable uncertainty estimates, which can help users and downstream systems make safer decisions by avoiding overconfidence.

There are two fundamental types of calibration: confidence calibration [23] and classwise calibration [15]. In this paper, we focus on both types of calibration.

Confidence calibration: Confidence calibration refers to the alignment of predicted confidence with observed frequencies. It ensures that a model's prediction confidence accurately reflect the true likelihood of an event occurring. A widely used metric for evaluating confidence calibration is the Expected Calibration Error (ECE) [23]. The definition of ECE can be found in Appendix A.

Classwise calibration: Classwise calibration is concerned with calibrating the predicted probabilities for each individual class in a multi-class classification problem. It ensures that the predicted probabilities across different classes are accurate and properly calibrated. The metric for classwise calibration is classwise-ECE (cw-ECE) [15], which can be found in Appendix A.

According to the number of models used for inference, calibration methods can be divided into single-model calibration methods and DE [5, 17].

Single-model calibration methods: Within single-model

calibration methods, there are two categories: train-time calibration methods and post-hoc calibration methods. Train-time calibration methods typically involve designing loss functions or incorporating additional regularization techniques, such as Entropy Regularization (ER) [26], Label Smoothing (LS) [22, 28], Focal Loss (FL) [21], Maximum Mean Calibration Error (MMCE) [16], Difference between Confidence and Accuracy (DCA) [19], Multi-class Difference in Confidence and Accuracy (MDCA) [10], and Dynamic Train-time Data Pruning (DTDP) [25]. Post-hoc calibration methods are applied after the model is trained, such as Temperature Scaling (TS) [27], Vector Scaling, and Dirichlet Scaling (DS) [15].

Deep Ensembles: DE creates an ensemble by training multiple models independently with different initializations. To alleviate overconfidence and improve calibration, during inference, DE leverages N randomly initialized, individually trained DNNs and makes predictions by average voting. While effective for enhancing model performance, DE introduces additional computational overhead during inference.

More related works are discussed in Appendix B.

3. Method

Let P_{data} denote the probability distribution of data. A dataset consists of finite (x, y_x) pairs where $x \sim P_{data}$ is a data instance and $y_x \in \mathcal{Y} = \{1, 2, \dots, K\}$ is the ground-truth class label of x . A model takes in the instance x and outputs its prediction vector $p_x \in \mathbb{R}^K$. Let $p_x^{(k)}$ denote the k -th dimension of p_x . The confidence $\hat{p}_x = \max_{k \in \mathcal{Y}} \{p_x^{(k)}\}$ is the maximum value in the probability vector of predictions, and the model's prediction $\hat{y}_x = \operatorname{argmax}_{k \in \mathcal{Y}} \{p_x^{(k)}\}$ is the index corresponding to the maximum value.

Typically, we train a model by minimizing the following CE loss on the training set,

$$L_{CE}(p_x) = -\log p_x^{(y_x)}, \quad (1)$$

where $p_x^{(y_x)}$ represents the prediction probability corresponding to the ground-truth label y_x . Previous studies [2, 8, 21] reveal that minimizing the CE loss forces the model to overfit to one-hot encoded ground-truth labels with almost 100% probability irrespective of their correctness, which can result in overconfidence.

DML [29] is a model distillation technique that allows two models $f(\theta)$ and $g(\phi)$ to learn collaboratively through a combination of a supervised loss and a Kullback-Leibler (KL) Divergence:

$$L_{f,x}(\theta) = L_{CE}(f_x(\theta)) + \alpha D_{KL}(g_x(\phi) || f_x(\theta)), \quad (2)$$

$$L_{g,x}(\phi) = L_{CE}(g_x(\phi)) + \alpha D_{KL}(f_x(\theta) || g_x(\phi)), \quad (3)$$

where α is a hyperparameter used to adjust the weight of the KL divergence loss.

In this paper, we designate $f(\theta)$ as the primary model, and $g(\phi)$ as the auxiliary model that does not participate in the inference process. It is important to note that DML was originally conceived as a knowledge distillation technique aimed at improving accuracy rather than focusing on calibration.

3.1. DML is an implicit form of deep ensembles

DML involves two models learning collaboratively by continuously exchanging knowledge during the training process. This collaborative learning encourages each model to learn from the strengths and correct the weaknesses of others, promoting diversity in model predictions. Consequently, DML achieves the advantages of DE—such as increased prediction accuracy and improved uncertainty calibration—without the need for independent model training or the associated computational costs.

In this section, we explore the theoretical relationship between DML and DE.

Given two trained models $f(\theta)$ and $g(\phi)$ and an input x for DE, we can get a weighted sum of predictive probabilities from $f_x(\theta)$ and $g_x(\phi)$ as the final prediction:

$$h_x(\theta, \phi) = (1 - \beta)f_x(\theta) + \beta g_x(\phi), \quad (4)$$

where $\beta \in [0, 1]$ is a weight to balance the effect of each model. To explain the relationship between DML and DE, we first present a proposition, the proof is in Appendix C).

Proposition 1. *Given two prediction probability distributions $f_x(\theta)$ and $g_x(\phi)$ for classifying sample x , as well as their ensemble $h_x(\underline{\theta}, \phi)$, we have*

$$\frac{\partial}{\partial \theta} D_{KL}(h_x(\underline{\theta}, \phi) || f_x(\theta)) = \beta \frac{\partial}{\partial \theta} D_{KL}(g_x(\phi) || f_x(\theta)). \quad (5)$$

Note that the variables with underscores do not participate in gradient propagation.

The left-hand side of (5) suggests that when we update θ to make f_x closer to h_x , we temporarily treat h_x as a model with fixed parameters, though, in the next step, h_x will also be updated with the new θ . By substituting (5) into (2) we obtain

$$L_{f,x}(\theta) = L_{CE}(f_x(\theta)) + \frac{\alpha}{\beta} D_{KL}(h_x(\underline{\theta}, \phi) || f_x(\theta)). \quad (6)$$

Similarly, the loss of model g can be rewritten as

$$L_{g,x}(\phi) = L_{CE}(g_x(\phi)) + \frac{\alpha}{1 - \beta} D_{KL}(h_x(\theta, \underline{\phi}) || g_x(\phi)). \quad (7)$$

From this perspective, we can see that KL divergence in DML helps continuously align each model's predictions with the ensemble distribution during training, thereby DML can be regarded as an implicit DE.

3.2. Differentiated Deep Mutual Learning

Model diversity plays a crucial role in the effectiveness of the ensemble [1, 30]. Although DML offers the opportunity to reduce computational overhead compared to DE, prior learning strategies in DML may destroy the model diversity and result in poor calibration performance. To address this issue, we propose an improved approach, Diff-DML, which increases the output discrepancy between f and g during training. Diff-DML employs two methods to increase model discrepancy: Differentiated Training Strategy (DTS) and Diversity-Preserving Learning Objective (DPLO).

Differentiated Training Strategy: In traditional DML training [29], both models typically use the same scheduler, beginning with a specific learning rate and gradually decreasing it over time. This consistency in training strategies often leads to high similarity between the two models, diminishing their diversity. A straightforward way to increase the diversity between the outputs of the two models is to employ different training schedulers in training.

In light of this, we propose to use different training strategies: one model, f , is designated as the primary model (used solely during inference) and follows the conventional strategy of gradually decreasing the learning rate. The other model, g , serves as an auxiliary model and is trained using a fixed learning rate throughout the training process. This discrepant learning rate scheme can preserve the diversity among models and thus improve the calibration performance.

Diversity-Preserving Learning Objective: To increase the disparity between the two models, we propose a Diversity-Preserving Learning Objective by changing the loss function of the auxiliary model in DML. In traditional DML, the models $f(\theta)$ and $g(\phi)$ share the same form of loss function, which combines a supervised loss with KL divergence. This learning objective may cause the model to overfit the one-hot ground-truth labels, resulting in unavoidable overconfidence and insufficient diversity. To address this problem, we suggest that the primary model f continue using the original loss function (2), while the auxiliary model $g(\phi)$ omits the cross-entropy loss component. As a result, the auxiliary model no longer receives direct supervision from the labels. Concretely, the new loss function of model $g(\phi)$ is

$$L_{g,x}(\phi) = D_{\text{KL}}(f_x(\theta) || g_x(\phi)). \quad (8)$$

The KL divergence not only constructs a mutual feedback loop between the primary and auxiliary models but also assists the auxiliary model in learning a differentiated prediction distribution to boost model diversity.

3.3. Extending to multiple auxiliary models

Naturally, we can incorporate multiple auxiliary models $\{g_i\}_{i=1}^{N_g}$ into the training process to further improve the prediction performance and calibration of the primary model f .

Analogous to DE, leveraging multiple auxiliary models can effectively promote model diversity and improve the uncertainty calibration of the primary model. The optimization objectives for the primary model f is

$$L_{f,x}(\theta) = L_{\text{CE}}(f_x(\theta)) + \alpha \sum_{i=1}^{N_g} \frac{1}{N_g} D_{\text{KL}}(g_{i,x}(\phi_i) || f_x(\theta)), \quad (9)$$

For traditional DML, the optimization objectives for each g_i are formalized as

$$L_{g_i,x}(\phi_i) = L_{\text{CE}}(g_{i,x}(\phi_i)) + \alpha D_{\text{KL}}(f_x(\theta) || g_{i,x}(\phi_i)), \quad (10)$$

where ϕ_i denotes the trainable parameters of the auxiliary model g_i . And for Diff-DML, the optimization objectives for each g_i are formalized as

$$L_{g_i,x}(\phi_i) = D_{\text{KL}}(f_x(\theta) || g_{i,x}(\phi_i)). \quad (11)$$

Similarly, we only use the primary model f in inference.

4. Experiments

We conducted primary experiments on image classification benchmarks, followed by experiments on misclassification and OOD detection to validate the performance of Diff-DML across different tasks.

4.1. Experimental setup

Datasets: We trained and evaluated our multi-class classification model using CIFAR-10 [13], CIFAR-100 [13] and Tiny-ImageNet [3] datasets. Validation sets were created by randomly selecting 5,000 samples from each dataset's training set, with the remaining samples used for training. We experimented with two network architectures, ResNet34 and ResNet50 [9], on these datasets.

Evaluation metrics: For standard classification, we evaluated all methods using accuracy (acc), ECE, and cw-ECE. For misclassification and OOD detection, we used four metrics: (1) False positive rate at 95% true positive rate (FPR-95%); (2) The misclassification probability when true positive rate is 95% (D-error); (3) Area under the receiver operating characteristic curve (AUROC); (4) Area under the precision-recall curve (AUPR).

Selection of models for testing: When selecting the optimal calibration model, we need to evaluate both accuracy and calibration performance. It is insufficient to consider only accuracy or only calibration performance. To account for both aspects simultaneously, we have developed a new method for selecting models to be tested on the test set, which we refer to as the Comprehensive Selection Method for Accuracy and Calibration (CSMAC). The selection method is as follows: A threshold value for accuracy, denoted as b , is set. When the highest accuracy of a model on the validation set is

below b , the checkpoint with the highest accuracy is selected. When the highest accuracy exceeds b , the checkpoint with the highest value of accuracy minus ECE (which we call the comprehensive evaluation score) on the validation set is selected. The value of b varies depending on the dataset and base model. In the experiment, we first train a model optimized using cross-entropy loss and identify the checkpoint corresponding to the highest accuracy on the validation set. We set this checkpoint as the baseline model, and denote its accuracy on the validation set as a . Then, we set $b = a - c$, where c represents a tolerance value for accuracy reduction, and in our experiment, c is set to 0.01. In this way, we establish a selection mechanism that takes both accuracy and calibration performance into account.

Competing methods and hyper-parameters: The evaluated methods include: **Baseline:** This is the model with the highest accuracy on the validation set trained with the CE loss. Note that the baseline model is used to determine the threshold b for the CSMAC selection method. All other models use this threshold to make selections through CSMAC. **Cross-Entropy (CE):** This is the model trained with CE loss (1) selected by CSMAC. **DML & DML-2:** These refer to the DML method using 1 and 2 auxiliary ResNet18 models, respectively. The primary model used was either ResNet34 or ResNet50. For both DML-1 and DML-2, we report the performance of the primary models. **Diff-DML & Diff-DML-2:** Similar to DML-1/2, these use Diff-DML for training with 1 or 2 auxiliary ResNet18 models. The performance of the primary model, either ResNet34 or ResNet50, is reported. For DML and Diff-DML, the value $\alpha \in [0.4, 1.0]$ and we tuned it to achieve the best performance via the validation set. **DCA [19]:** Four values were used for its β : 0.5, 1.0, 5.0, and 10.0. **MMCE [16]:** Four values were used for its λ : 1.0, 2.0, 3.0, and 4.0. **FL [21]:** Three values were used for its γ : 1.0, 2.0, and 3.0. **FL+MDCA [10]:** Three values were used for the γ in the focal loss: 1.0, 2.0, and 3.0, while the β value in MDCA was 1.0. **FL+DTDP [25]:** Three values were used for the γ in the focal loss: 1.0, 2.0, and 3.0. For DTDP, we used an α value of 0.005 and experimented with λ values of 5.0 and 10.0, along with a data pruning ratio of 10%. **DE-3 [5, 17]:** It denotes DE with 3 models using cross-entropy loss with different initializations and combined their produced probabilities.

Other implementation details: In the experiments, we utilized the SGD optimizer with a momentum of 0.9 and weight decay of 0.0005. The initial learning rate was set to 0.1 for all datasets. For each dataset, we trained the models for a total of 250 epochs. During the initial 100 epochs, we maintained the initial learning rate. Afterward, every 50 epochs, we decreased the learning rate by a factor of 10%. In Diff-DML, the primary model f employed the same optimization strategy as described above. However, for the

auxiliary model g , we initialized the learning rate as 0.01 and did not apply any learning rate decay. All experiments were performed with a batch size of 100 and conducted for training and testing on a single GeForce RTX 2080 GPU.

4.2. Classification accuracy and calibration

Table 1 shows the comparison results of Diff-DML and various single-model calibration methods on CIFAR-100. Tables S1 and S2 in Appendix D show the comparison results on CIFAR-10 and Tiny-ImageNet, respectively. Figure S1 in Appendix E visually demonstrates the calibration performance of Diff-DML with reliability diagrams and confidence histograms.

Comparison between DML and Diff-DML: From Tables 1, S1 and S2, it can be observed that on all three datasets, DML and Diff-DML achieved improvements in both accuracy and calibration performance compared to CE. The improvements of Diff-DML were more significant than those of DML.

Figure 2 shows the validation accuracy and ECE curves of ResNet34 during training on the CIFAR-100 dataset using CE, DML and Diff-DML methods. It can be seen that the ECE values for CE and DML steadily increased throughout training, while the ECE values for Diff-DML showed a downward trend.

Comparison with regularization methods: It can be observed that the majority of regularization methods compromised a certain level of accuracy to improve calibration ability. For instance, a previous state-of-the-art method, FL+MDCA, reduced accuracy by 1 to 2 percents. In contrast, our Diff-DML method avoided this trade-off and demonstrates superior accuracy and calibration ability.

In terms of calibration performance, Diff-DML surpassed FL+MDCA across all three datasets. Specifically, compared to FL+MDCA, Diff-DML achieved an absolute accuracy increase of 1.3%/3.1% for ResNet34/ResNet50 on the CIFAR-100 dataset, with relative reductions in ECE of 49.6%/43.8% and relative reductions in cw-ECE of 7.7%/13.0%, representing a considerable improvement.

Comparison with post-hoc calibration methods: Post-hoc calibration methods in general did not affect accuracy much, but their calibration performance on datasets with many classes, like CIFAR-100 and Tiny-ImageNet, was often suboptimal. In contrast, Diff-DML showed much better calibration performance on these datasets.

Results with more auxiliary models: Table 2 shows the results of DML and Diff-DML trained with more auxiliary models on the CIFAR-100 dataset, compared to DE-3. Table S3 in Appendix D presents the comparison on CIFAR-10 and Tiny-ImageNet. It can be observed that DE-3 provided the most significant improvement in accuracy. DML and Diff-DML with two auxiliary models slightly outperformed those with one auxiliary model, but the difference was not substantial. In terms of calibration performance, Diff-DML

Method	Acc(%) $\uparrow$		ECE(%) $\downarrow$		cw-ECE(10^{-3}) $\downarrow$	
	ResNet34	ResNet50	ResNet34	ResNet50	ResNet34	ResNet50
Baseline	77.87	77.85	8.81	10.39	2.25	2.51
CE	77.73	77.54	8.55	10.04	2.29	2.44
<i>Post-hoc calibration methods</i>						
CE+TS [8]	77.73	77.54	3.37	3.14	1.72	1.63
CE+VS [8]	77.47	77.33	3.76	3.13	1.82	1.71
CE+DS [15]	77.33	77.54	3.20	3.14	1.71	1.63
<i>Regularization based calibration methods</i>						
DCA [19]	77.53	77.66	10.10	11.69	2.43	2.66
MMCE [16]	75.50	75.16	6.26	7.19	1.85	1.98
FL [21]	76.93	76.49	1.21	1.79	1.56	1.55
FL+DTDP [25]	75.87	76.09	1.82	2.61	1.60	1.65
FL+MDCA [10]	76.91	75.93	1.29	1.53	1.55	1.61
<i>DML-based calibration methods</i>						
DML [29]	77.36	77.89	3.43	4.42	1.56	1.69
Diff-DML (ours)	78.20	79.05	0.65	0.86	1.43	1.40

Table 1. Different calibration methods' accuracy, ECE, and cw-ECE on the CIFAR-100 dataset.

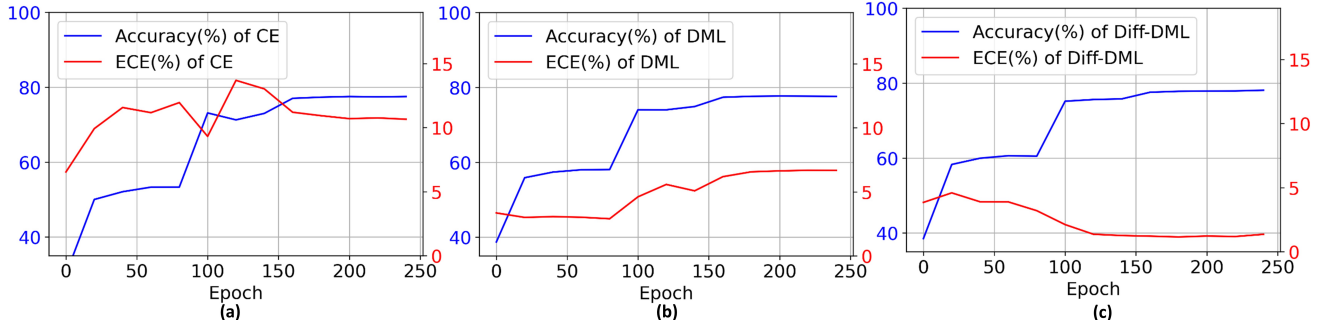


Figure 2. The validation accuracy and ECE curves of ResNet34 trained using the (a) CE, (b) DML, and (c) Diff-DML methods on the CIFAR-100 dataset during the training process.

Method	Acc(%) $\uparrow$		ECE(%) $\downarrow$		cw-ECE(10^{-3}) $\downarrow$	
	ResNet34	ResNet50	ResNet34	ResNet50	ResNet34	ResNet50
DE-3 [5, 17]	79.34	79.63	2.95	4.40	1.56	1.73
DML [29]	77.36	77.89	3.43	4.42	1.56	1.69
DML-2 [29]	77.98	78.00	4.32	4.38	1.61	1.82
Diff-DML (ours)	78.20	79.05	0.65	0.86	1.43	1.40
Diff-DML-2 (ours)	79.14	79.30	0.75	0.72	1.37	1.33

Table 2. Comparison of DML and Diff-DML trained with more auxiliary models against DE on CIFAR-100 dataset.

was clearly superior to DE-3.

Training and inference time: In terms of time overhead, training Diff-DML with one auxiliary model took 1.4 times longer than training a single model, and with two auxiliary models, it took 1.9 times longer. In contrast, DE-3's training time was 3 times that of a single model. For inference, Diff-DML required the same time as a single model, while

DE-3's inference time was 3 times longer. Thus, our method provided substantial advantages over DE-3 in both training and inference time.

4.3. Analysis of the robustness of Diff-DML

Sensitivity of α : We investigated the sensitivity of the important hyper-parameter α in (2). From Figure 3, it is ob-

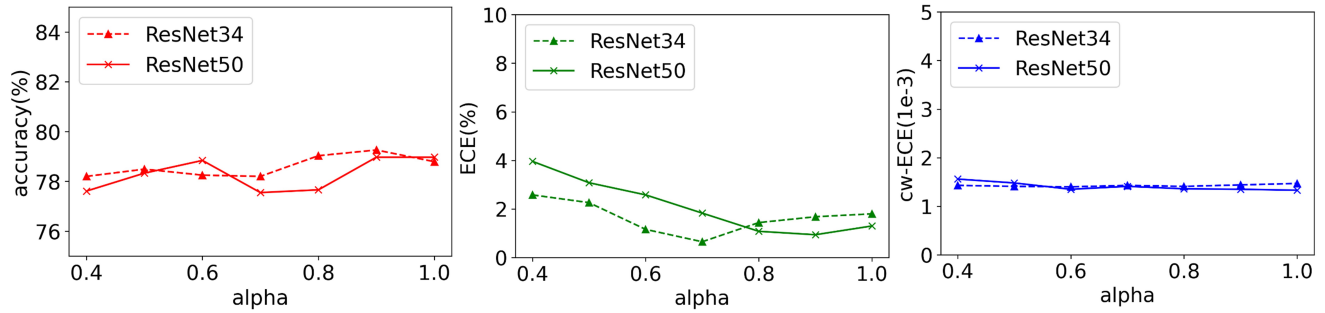


Figure 3. Variation of accuracy, ECE, and classwise-ECE of Diff-DML method trained on CIFAR-100 with different values of hyper-parameter α .

served that on CIFAR-100, as α changes, both accuracy and cw-ECE exhibited smooth variations. Specifically, the fluctuations in accuracy remained within 3 percentage points, while the fluctuations in cw-ECE were within 0.02 percentage points. As for ECE, although its values ranged between 2 and 4 for very small (e.g., 0.4) and large (e.g., 1.0) α values, the fluctuations around the lowest point of ECE were also minimal. This demonstrates the good robustness of Diff-DML with respect to the hyper-parameter α .

Selection of auxiliary model: We investigated how to select the auxiliary model. We experimented with CNNs containing two hidden layers, as well as ResNet18, ResNet34, and ResNet50 as backbones for the auxiliary model, which is shown in Table S4 in Appendix F. It can be observed that only when using CNNs, both accuracy and calibration ability were significantly poor. When using ResNet18, ResNet34, and ResNet50, similar performance were observed. This indicates that in order to ensure good performance of the Diff-DML, a backbone for the auxiliary model should not be selected if its performance is significantly worse than that of the primary model. Furthermore, when performance is similar, specific model architectures will yield similar performance.

4.4. Ablation study

In Table 3, we conducted an ablation study on the performance of the two strategies, DTS and DPLO, used in Diff-DML, using ResNet50 on the CIFAR-100 dataset. In the ablation study of DPLO, we set the learning rates and schedulers of models f and g to be the same as those in DML. As shown in the table, Diff-DML with only DTS or only DPLO both achieve better accuracy and calibration performance compared to the original DML, with the best results obtained when both strategies are combined.

4.5. Misclassification & OOD detection

A real-world application of calibrated predictions is making the model aware of its uncertainties. Specifically, the model should show low confidence for misclassified or out-of-distribution samples. Motivated by this, we conducted

Method	Acc(%) $\uparrow$	ECE(%) $\downarrow$	cw-ECE(%) $\downarrow$
CE	77.54	10.04	2.44
DML [29]	77.89	4.42	1.69
DTS	78.10	1.38	1.43
DPLO	78.75	1.01	1.40
DTS+DPLO	79.05	0.86	1.40

Table 3. Ablation study of Diff-DML on CIFAR-100.

experiments on misclassification and OOD detection, measuring the success rate of detecting such samples using maximum softmax probability [11]. We report threshold-free metrics (FPR-95%, D-error, AUROC, and AUPR).

Misclassification detection: Misclassification detection is an important problem in machine learning, as it allows for the identification of instances where the model’s predictions are unreliable. We expect the model to output low probability or high uncertainty for samples that cannot be classified correctly. Table 4 presents the performance results for various models in detecting misclassifications. Diff-DML showed significant improvements over other single-model calibration techniques and the DE method.

OOD detection: A reliable classification model should show higher uncertainty and lower confidence when encountering test samples that differ significantly from the training data. We evaluated the ability of different calibration methods to distinguish OOD samples by mixing in-distribution test data with OOD data. We tested models trained on CIFAR-10 and CIFAR-100 for two types of OOD detection [5, 12]: far OOD (distinguishing SVHN [24] from CIFAR-10/CIFAR-100) and near OOD (distinguishing between CIFAR-10 and CIFAR-100). The results in Table 5 show that Diff-DML significantly outperformed other calibration methods, even exceeding the DE method, which is known for its OOD detection effectiveness.

5. Conclusion and discussion

Although Diff-DML significantly enhances both accuracy and calibration, its reliance on an auxiliary model with performance comparable to the primary model presents challenges, especially when training models with large parameter

Dataset	Method	FPR-95%↓	D-error↓	AUROC↑	AUPR↑
CIFAR-10	CE	38.53	13.20	91.80	99.46
	DCA [19]	38.97	13.28	91.94	99.47
	MMCE [16]	23.00	12.61	93.59	99.54
	FL [21]	33.14	13.56	92.23	99.46
	FL+DTDP [25]	28.34	14.26	92.20	99.36
	FL+MDCA [10]	32.80	13.21	92.41	99.47
	DE-3 [5, 17]	22.37	10.89	94.08	99.68
	DML [29]	19.22	11.49	94.05	99.64
	Diff-DML(ours)	16.63	10.00	94.87	99.73
CIFAR-100	CE	45.95	20.19	86.93	95.47
	DCA [19]	40.66	20.09	86.85	95.01
	MMCE [16]	45.97	20.02	87.00	95.05
	FL [21]	45.02	20.21	87.04	95.42
	FL+DTDP [25]	43.95	20.73	86.79	95.16
	FL+MDCA [10]	45.08	20.68	86.56	95.36
	DE-3 [5, 17]	40.04	19.25	88.19	96.44
	DML [29]	40.31	19.42	87.87	96.17
	Diff-DML(ours)	39.62	18.95	88.38	96.50

Table 4. The misclassification detection performance of different calibration methods.

Dataset	Method	FPR-95%↓	D-error↓	AUROC↑	AUPR↑
In-Distribution: CIFAR-10	CE	69.48/19.80	19.45/10.59	86.11/94.60	82.00/72.54
	DCA [19]	64.63/18.38	18.80/10.10	87.07/95.05	84.36/72.48
	MMCE [16]	40.40/11.93	17.33/8.37	89.13/ 97.53	89.61/88.85
	FL [21]	66.13/19.20	20.97/10.69	85.57/94.28	82.81/74.18
	FL+DTDP [25]	50.54/23.62	20.38/13.54	86.57/92.03	86.73/77.10
	FL+MDCA [10]	72.19/29.84	20.75/13.58	84.77/92.63	81.01/67.11
	DE-3 [5, 17]	33.89/13.31	15.60/9.07	90.12/95.06	90.27/86.53
	DML [29]	32.42/13.85	15.90/9.33	91.05/95.38	91.67/86.04
	Diff-DML(ours)	30.12/10.51	15.06/7.65	91.87/96.33	92.67/90.29
Out-of-Distribution: CIFAR-100/SVHN	CE	68.26/45.87	28.69/22.27	77.30/85.44	77.49/54.01
	DCA [19]	69.41/47.88	30.58/23.62	75.96/83.20	77.19/55.41
	MMCE [16]	65.34/49.38	29.53/25.77	76.58/77.64	78.66/49.53
	FL [21]	67.27/48.40	28.10/23.17	78.36/83.89	78.87/53.39
	FL+DTDP [25]	59.10/53.85	27.23/24.16	79.07/83.10	81.05/49.61
	FL+MDCA [10]	69.52/41.66	27.93/19.87	78.35/ 87.46	77.92/59.08
	DE-3 [5, 17]	56.21 /48.70	26.61/24.20	80.36/83.42	82.34/53.17
	DML [29]	62.08/37.95	27.07/19.68	79.90/87.06	81.07/65.32
	Diff-DML(ours)	56.21/37.84	25.55/19.58	81.50/87.34	83.57/65.65

Table 5. The OOD detection performance of different calibration methods. The numbers before and after the backslash denote the near and far OOD detection results, respectively.

sizes. This dependency can complicate scalability and computational efficiency, particularly in resource-constrained environments.

In conclusion, we introduce Diff-DML, which overcomes the limitations of traditional DML by incorporating two strategies: DTS and DPLO. While DML promotes learning from an ensemble distribution, it often reduces prediction diversity, which is crucial for effective calibration. Diff-DML preserves this diversity while simultaneously

improving both accuracy and calibration. Extensive experiments on datasets such as CIFAR-100 demonstrate that our approach outperforms state-of-the-art methods, offering a practical and efficient solution for reliable uncertainty estimation, particularly in safety-critical applications.

References

- [1] Gavin Brown. *Diversity in neural network ensembles*. PhD thesis, Citeseer, 2004. 4
- [2] Peng Cui, Dan Zhang, Zhijie Deng, Yinpeng Dong, and Jun Zhu. Learning sample difficulty from pre-trained models for reliable prediction. *Advances in Neural Information Processing Systems*, 36, 2024. 2, 3
- [3] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *2009 IEEE conference on computer vision and pattern recognition*, pages 248–255. Ieee, 2009. 4
- [4] James M Dolezal, Andrew Srisuwananukorn, Dmitry Karpeyev, Siddhi Ramesh, Sara Kochanny, Brittany Cody, Aaron S Mansfield, Sagar Rakshit, Radhika Bansal, Melanie C Bois, et al. Uncertainty-informed deep learning models enable high-confidence predictions for digital histopathology. *Nature communications*, 13(1):6572, 2022. 1
- [5] Stanislav Fort, Huiyi Hu, and Balaji Lakshminarayanan. Deep ensembles: A loss landscape perspective. *arXiv preprint arXiv:1912.02757*, 2019. 2, 5, 6, 7, 8, 1
- [6] Kevin P Greenman, Ava P Amini, and Kevin K Yang. Benchmarking uncertainty quantification for protein engineering. *bioRxiv*, pages 2023–04, 2023. 1
- [7] Arthur Gretton. Introduction to rkhs, and some simple kernel algorithms. *Adv. Top. Mach. Learn. Lecture Conducted from University College London*, 16(5-3):2, 2013. 1
- [8] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q Weinberger. On calibration of modern neural networks. In *International conference on machine learning*, pages 1321–1330. PMLR, 2017. 2, 3, 6
- [9] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 770–778, 2016. 4
- [10] Ramya Hebbalaguppe, Jatin Prakash, Neelabh Madan, and Chetan Arora. A stitch in time saves nine: A train-time regularizing loss for improved neural network calibration. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 16081–16090, 2022. 2, 3, 5, 6, 8, 1
- [11] Dan Hendrycks and Kevin Gimpel. A baseline for detecting misclassified and out-of-distribution examples in neural networks. In *International Conference on Learning Representations*, 2016. 7
- [12] Dan Hendrycks, Steven Basart, Mantas Mazeika, Andy Zou, Joe Kwon, Mohammadreza Mostajabi, Jacob Steinhardt, and Dawn Song. Scaling out-of-distribution detection for real-world settings. *arXiv preprint arXiv:1911.11132*, 2019. 7
- [13] Alex Krizhevsky, Geoffrey Hinton, et al. Learning multiple layers of features from tiny images. 2009. 4
- [14] Meelis Kull, Telmo Silva Filho, and Peter Flach. Beta calibration: a well-founded and easily implemented improvement on logistic calibration for binary classifiers. In *Artificial Intelligence and Statistics*, pages 623–631. PMLR, 2017. 1
- [15] Meelis Kull, Miquel Perello Nieto, Markus Kängsepp, Telmo Silva Filho, Hao Song, and Peter Flach. Beyond temperature scaling: Obtaining well-calibrated multi-class probabilities with dirichlet calibration. *Advances in neural information processing systems*, 32, 2019. 2, 3, 6, 1
- [16] Aviral Kumar, Sunita Sarawagi, and Ujjwal Jain. Trainable calibration measures for neural networks from kernel mean embeddings. In *International Conference on Machine Learning*, pages 2805–2814. PMLR, 2018. 2, 3, 5, 6, 8, 1
- [17] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. *Advances in neural information processing systems*, 30, 2017. 2, 5, 6, 8, 1
- [18] Christian Leibig, Vaneeda Allken, Murat Seçkin Ayhan, Philipp Berens, and Siegfried Wahl. Leveraging uncertainty information from deep neural networks for disease detection. *Scientific reports*, 7(1):1–14, 2017. 1
- [19] Gongbo Liang, Yu Zhang, Xiaoqin Wang, and Nathan Jacobs. Improved trainable calibration method for neural networks on medical imaging classification. *arXiv preprint arXiv:2009.04057*, 2020. 3, 5, 6, 8, 1
- [20] Rhiannon Michelmore, Matthew Wicker, Luca Laurenti, Luca Cardelli, Yarin Gal, and Marta Kwiatkowska. Uncertainty quantification with statistical guarantees in end-to-end autonomous driving control. In *International Conference on Robotics and Automation*, 2020. 1
- [21] Jishnu Mukhoti, Viveka Kulharia, Amartya Sanyal, Stuart Golodetz, Philip Torr, and Puneet Dokania. Calibrating deep neural networks using focal loss. *Advances in Neural Information Processing Systems*, 33:15288–15299, 2020. 3, 5, 6, 8, 1
- [22] Rafael Müller, Simon Kornblith, and Geoffrey E Hinton. When does label smoothing help? *Advances in neural information processing systems*, 32, 2019. 2, 3, 1
- [23] Mahdi Pakdaman Naeini, Gregory Cooper, and Milos Hauskrecht. Obtaining well calibrated probabilities using bayesian binning. In *Proceedings of the AAAI conference on artificial intelligence*, 2015. 2
- [24] Yuval Netzer, Tao Wang, Adam Coates, Alessandro Bissacco, Baolin Wu, Andrew Y Ng, et al. Reading digits in natural images with unsupervised feature learning. In *NIPS workshop on deep learning and unsupervised feature learning*, page 7. Granada, Spain, 2011. 7
- [25] Rishabh Patra, Ramya Hebbalaguppe, Tirtharaj Dash, Gautam Shroff, and Lovekesh Vig. Calibrating deep neural networks using explicit regularisation and dynamic data pruning. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 1541–1549, 2023. 3, 5, 6, 8, 1
- [26] Gabriel Pereyra, George Tucker, Jan Chorowski, Łukasz Kaiser, and Geoffrey Hinton. Regularizing neural networks by penalizing confident output distributions. *arXiv preprint arXiv:1701.06548*, 2017. 2, 3, 1
- [27] John Platt et al. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. *Advances in large margin classifiers*, 10(3):61–74, 1999. 2, 3, 1
- [28] Christian Szegedy, Vincent Vanhoucke, Sergey Ioffe, Jon Shlens, and Zbigniew Wojna. Rethinking the inception architecture for computer vision. In *Proceedings of the IEEE*

- conference on computer vision and pattern recognition*, pages 2818–2826, 2016. [2](#), [3](#), [1](#)
- [29] Ying Zhang, Tao Xiang, Timothy M Hospedales, and Huchuan Lu. Deep mutual learning. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 4320–4328, 2018. [2](#), [3](#), [4](#), [6](#), [7](#), [8](#)
- [30] Zhi-Hua Zhou. *Ensemble methods: foundations and algorithms*. CRC press, 2012. [4](#)